SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	V.SHASHANK SAI	**Reg. No.:**	192524509
Date:	8-10-2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders** ○ Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories** ○ Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria** ○ Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog** ○ Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog** ○ Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points** ○ Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables** ○ Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

AI-Powered Crop Yield Prediction and Optimization

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

The vision of the AI-Powered Crop Yield Prediction and Optimization system is to develop a data-driven agricultural decision-support platform that can estimate crop yield at an early stage and provide useful recommendations for crop planning and resource optimization. The system analyzes factors such as crop type, soil characteristics, rainfall, temperature, irrigation, fertilizer usage, and historical production to generate yield predictions and support betterinformed farming decisions.

Objectives

- Predict expected crop yield using relevant agricultural and environmental data.
- Identify important factors that influence crop productivity.
- Support informed crop selection and seasonal farm planning.
- Provide data-driven recommendations for efficient use of water, fertilizer, and other inputs.
- Present predictions, trends, and recommendations through an accessible dashboard.
- Improve planning by reducing dependence on manual estimation alone. **Key Stakeholders**

Stakeholder	Role
Farmers	Provide farm information and use predictions for crop and resource planning.
Agricultural Experts / Extension Officers	Review recommendations and interpret results according to local farming conditions.
Agricultural Researchers	Analyze agricultural patterns and evaluate model usefulness.
Data Scientists / ML Engineers	Prepare data, build models, evaluate performance, and improve predictions.
Software Developers	Develop, test, integrate, and maintain the application.
System Administrators	Manage users, permissions, database, backups, and system availability.
Agricultural Organizations / Government Departments	Use aggregated insights for agricultural planning and advisory services.

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

Epic 1 – Agricultural Data Management

User Story 1:

As a farmer, I want to enter and update crop, soil, weather, and farm information, so that the system has current data for analysis.

User Story 2:

As an administrator, I want to validate uploaded agricultural records, so that incomplete or incorrect information does not affect predictions.

Epic 2 – Data Analysis and Preparation

User Story 3:

As a data analyst, I want to clean and preprocess historical agricultural data, so that it can be used reliably for model training.

User Story 4:

As an agricultural researcher, I want to analyze relationships between soil, weather, inputs, and yield, so that important productivity factors can be identified. **Epic 3 – Crop Yield Prediction and Optimization**

User Story 5:

As a farmer, I want the system to predict expected crop yield, so that I can plan production before harvest.

User Story 6:

As a farmer, I want recommendations for crop and resource planning, so that I can improve productivity and use inputs efficiently.

Epic 4 – Dashboard and Reporting

User Story 7:

As a farmer or agricultural expert, I want to view predicted yield and related trends on a dashboard, so that I can understand the results easily.

User Story 8:

As an administrator or researcher, I want to generate reports on predictions and agricultural factors, so that I can support planning and evaluation.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1 – Enter Agricultural Details

Given the farmer is logged into the system,
When the farmer enters valid crop, soil, and farm details and submits the form, Then
the system validates and stores the information successfully.

User Story 3 – Prepare Historical Data

Given historical agricultural records are available,
When the analyst runs the preprocessing process,
Then the system produces a cleaned dataset suitable for model analysis.

User Story 5 – Predict Crop Yield

Given valid agricultural input data is available,
When the user requests a yield prediction,
Then the system calculates and displays the predicted yield.

User Story 6 – Optimization Recommendation Given

sufficient crop and farm information is available,

When the user requests recommendations,
Then the system displays relevant data-informed suggestions.

User Story 7 – Dashboard

Given the user has dashboard access,
When the dashboard is opened,
Then it displays prediction results, relevant trends, and key agricultural inputs.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

The backlog is prioritized using business value, user importance, dependencies, risk, and implementation feasibility.

Priority	User Story	Business Value	Reason
1	Enter and update agricultural data	Very High	Required for all other functions.
2	Validate agricultural records	Very High	Ensures reliable input data.
3	Prepare historical data	High	Required for model development.
4	Analyze productivity factors	High	Helps understand factors affecting yield.
5	Predict crop yield	Very High	Core feature of the system.
6	Generate optimization recommendations	Very High	Supports better farming decisions.
7	Dashboard	High	Provides easy monitoring and interpretation.
8	Reports	Medium	Supports documentation and decision-making.

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

Sprint 1 – Agricultural Data Management and Prediction Preparation

For the first sprint, the team selects the highest-priority stories that form the foundation for reliable crop-yield prediction.

User Story	Development Tasks
Enter and update agricultural data	Design data fields, create input form, implement database storage.
Validate agricultural records	Define validation rules, implement validation, display error messages.
Prepare historical data	Collect sample dataset, clean data, identify important attributes.
Analyze productivity factors	Analyze historical records, identify yield indicators, prepare model features.

Actionable Development Tasks

1. Design the agricultural data database.
2. Create the crop and farm information input form.
3. Implement input validation.
4. Store agricultural records in the database.
5. Prepare historical/sample datasets.
6. Clean and preprocess the dataset.
7. Identify important yield-related factors.

8. Prepare the dataset for the prediction model.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

The Fibonacci sequence (1, 2, 3, 5, 8, 13) is used for estimation. Higher points indicate greater complexity, uncertainty, and development effort.

User Story	Story Points	Justification
Enter and update agricultural data	5	Requires UI, database, validation, and CRUD operations.
Validate agricultural records	3	Requires validation rules and error handling.
Prepare historical data	5	Requires data collection, cleaning, and transformation.
Analyze productivity factors	5	Requires analysis of multiple agricultural parameters.
Predict crop yield	8	Involves developing, evaluating, and integrating a prediction model.
Optimization recommendations	5	Requires recommendation logic based on agricultural factors.
Dashboard	5	Requires data visualization and integration.
Reports	3	Requires report generation and formatting.

Total Estimated Story Points 39
story points

For the first sprint, the team can select approximately 13–18 points, depending on its actual capacity.

7. Present the Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Sprint Goal

"To establish a reliable agricultural data management and analysis foundation that prepares the system for accurate crop-yield prediction and optimization."

Expected Sprint Deliverables

At the end of Sprint 1, the team is expected to deliver:

1. **Agricultural Data Entry Module**
    - Add and update crop, soil, weather, and farm records. ○ Store data securely in the database.
2. **Data Validation**
    - Validate required fields.
    - Detect incomplete or invalid information.
3. **Historical Data Preparation**
    - Collect and organize sample/historical agricultural data.

- Clean and preprocess the dataset.

4. **Productivity Factor Analysis**
    - Identify soil, weather, irrigation, fertilizer, and other important factors.
5. **Prediction Preparation**
    - Prepare relevant features and datasets for the crop-yield prediction model.

Final Expected Outcome

The completed project will provide an AI-powered agricultural decision-support dashboard that can analyze crop, soil, weather, and farm information, estimate expected crop yield, identify important productivity factors, and provide data-informed optimization recommendations so that farmers and agricultural stakeholders can make better planning decisions.